JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

Functions of One Real Variable (2005–2026 Archive)

Category: Real Analysis • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
155	329 M	465 Min	MCQ • MSQ • NAT (2005–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	85	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	27	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	43	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

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Question 1 (JAM 2026 • Q54 • ID: JAM_2026_Q54) NAT +2 M Neg: Nil (0)

Q.54	Let $$f(x) = \int_0^{e^{\sin x}} \int_0^{\log_e y} e^{-\left(\frac{1}{\sqrt{2}}\right)} (1 - t^2)\, dt\, dy.$$ Then $f'(\pi/4) =$ _____(*rounded off to one decimal place*).

Question 2 (JAM 2026 • Q48 • ID: JAM_2026_Q48) NAT +1 M Neg: Nil (0)

Q.48	$\int_0^3 (\,|x-1| - x[x]\,)\, dx =$ _____(*rounded off to one decimal place*).

Question 3 (JAM 2026 • Q44 • ID: JAM_2026_Q44) NAT +1 M Neg: Nil (0)

Q.44	$\lim_{x\to 0} \left(\frac{1}{x} - \frac{1}{\sin x} + e^{\frac{1-\cos x}{x}} \right) =$ _____(*rounded off to one decimal place*).

Question 4 (JAM 2026 • Q34 • ID: JAM_2026_Q34) MSQ +2 M Neg: Nil (0)

Q.34	Let $f: \mathbb{R} \to \mathbb{R}$ be the function given by $$f(x) = \begin{cases} \cos x, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q}. \end{cases}$$ Which of the following statements is/are TRUE?
(A)	$f(x)$ is continuous at 0.
(B)	$f(x)$ is continuous at $\frac{\pi}{2}$.
(C)	$f(x)$ is Riemann integrable on [0, 1] and $\int_0^1 f(x)\,dx = \sin 1$.
(D)	$f(x)$ is Riemann integrable on [0, 1] and $\int_0^1 f(x)\,dx = 0$.

Question 5 (JAM 2026 • Q33 • ID: JAM_2026_Q33) MSQ +2 M Neg: Nil (0)

Q.33	Which of the following functions is/are differentiable at $x = 1$?
(A)	$f(x) = \lvert x-1\rvert^3$
(B)	$f(x) = \lvert x^2-1\rvert$
(C)	$f(x) = \begin{cases} x^2 e^{-x^2}, & \lvert x\rvert \le 1 \\ e^{-1}, & \lvert x\rvert > 1 \end{cases}$
(D)	$f(x) = [x]$

Question 6 (JAM 2026 • Q31 • ID: JAM_2026_Q31) MSQ +2 M Neg: Nil (0)

Q.31	Which of the following functions $f: \mathbb{R} \to \mathbb{R}$ has/have a local minimum at $x = 0$?
(A)	$f(x) = \sin \lvert x\rvert$
(B)	$f(x) = \sin x + \frac{x^3}{6}$
(C)	$f(x) = x^4 + x^2 + 3$
(D)	$f(x) = \min\{x - [x], 1 - x + [x]\}$

Question 7 (JAM 2026 • Q21 • ID: JAM_2026_Q21) MCQ +2 M Neg: −0.67

Q.21	Let $f:[1,9]\to\mathbb{R}$ be a non-constant continuous function such that $f(1)=f(9)$. Which ONE of the following statements is necessarily TRUE?
(A)	There exists $c\in[4,8]$ such that $f(c)=f(c+1)$.
(B)	There exists $c\in[3,7]$ such that $f(c)=f(c+2)$.
(C)	There exists $c\in[2,6]$ such that $f(c)=f(c+3)$.
(D)	There exists $c\in[1,5]$ such that $f(c)=f(c+4)$.

JAM 2026

Question 8 (JAM 2026 • Q19 • ID: JAM_2026_Q19) MCQ +2 M Neg: −0.67

Q.19	The value of the following expression $$\lim_{x\to\infty}\int_0^x e^{-t}\lvert\sin t\rvert\, dt$$ is _____.
(A)	$\dfrac{e^{\pi}+1}{2(e^{\pi}-1)}$
(B)	$\dfrac{e^{\pi}-1}{2(e^{\pi}+1)}$
(C)	$\dfrac{2(e^{\pi}+1)}{e^{\pi}-1}$
(D)	$\dfrac{2(e^{\pi}-1)}{e^{\pi}+1}$

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Question 9 (JAM 2026 • Q13 • ID: JAM_2026_Q13) MCQ +2 M Neg: −0.67

Q.13	Let $f:\mathbb{R}\to\mathbb{R}$ be a function that has derivatives of all orders. Which ONE of the following statements is FALSE?
(A)	If f is a polynomial in x of degree at most 5, then $f^{(6)}(x)=0$ for all $x\in\mathbb{R}$.
(B)	If $f^{(6)}(x)=0$ for all $x\in\mathbb{R}$, then f is a polynomial in x of degree at most 5.
(C)	If $f^{(k)}(a)=0$ for all $1\le k\le 5$ and $f^{(6)}(a)>0$ for some $a\in\mathbb{R}$, then $f(x)$ has a local minimum at a.
(D)	If $f^{(k)}(a)=0$ for all $1\le k\le 6$ and $f^{(7)}(a)<0$ for some $a\in\mathbb{R}$, then $f(x)$ has a local maximum at a.

Question 10 (JAM 2026 • Q7 • ID: JAM_2026_Q7) MCQ +1 M Neg: −0.33

Q.7	Let $f:\mathbb{R}\to\mathbb{R}$ be a twice differentiable function such that $f(n)=1$ for all $n\in\mathbb{Z}$. Which ONE of the following statements is necessarily TRUE?
(A)	$f'(n)=0$ for all $n\in\mathbb{Z}$.
(B)	$f'(x)=0$ for infinitely many $x\in\mathbb{R}\setminus\mathbb{Z}$.
(C)	$f''(n)=0$ for all $n\in\mathbb{Z}$.
(D)	$f''(x)=0$ for infinitely many $x\in\mathbb{R}\setminus\mathbb{Z}$.

Question 11 (JAM 2026 • Q6 • ID: JAM_2026_Q6) MCQ +1 M Neg: −0.33

Q.6	Let $f:(1,2)\to\mathbb{R}$ be a continuous function satisfying $\lim_{x\to 1+} f(x)=3$ and $\lim_{x\to 2-} f(x)=3$. Then which ONE of the following is necessarily TRUE?
(A)	f is bounded and f has a maximum or a minimum but not both.
(B)	f is bounded and f has a maximum or a minimum or both.
(C)	f is bounded and f has neither a maximum nor a minimum.
(D)	f is unbounded and f has either no maximum or no minimum.

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Question 12 (JAM 2026 • Q3 • ID: JAM_2026_Q3) MCQ +1 M Neg: −0.33

Q.3	Let $f:(1,4)\to\mathbb{R}$ be a differentiable function such that $$f'(x)=(f(x)-\pi x)^2+\pi \quad \text{for all } x\in(1,4).$$ Which ONE of the following is a possible value of $f(3)-f(2)$?
(A)	$\pi+\frac{1}{6}$
(B)	$\pi-\frac{1}{6}$
(C)	$\frac{\pi}{2}+\frac{1}{3}$
(D)	$\frac{\pi}{2}+\frac{1}{2}$

Question 13 (JAM 2025 • Q53 • ID: JAM_2025_Q53) NAT +2 M Neg: Nil (0)

Q.53 Let $f(x)=2x-\sin x$, for all $x\in\mathbb{R}$. Let $k\in\mathbb{N}$ be such that

$$\lim_{x\to 0}\left(\frac{1}{x}\sum_{i=1}^{k} i^2 f\left(\frac{x}{i}\right)\right)=45.$$

Then, the value of k is equal to _____.

Question 14 (JAM 2025 • Q51 • ID: JAM_2025_Q51) NAT +2 M Neg: Nil (0)

Q.51 Let $f: \mathbb{R} \to \mathbb{R}$ be a continuous function satisfying

$$\int_0^{\pi/4} \left(\sin(x)\, f(x) + \cos(x) \int_0^x f(t)dt \right) dx = \sqrt{2}\,.$$

Then, the value of $\int_0^{\pi/4} f(x)\, dx$ is equal to _____ *(rounded off to two decimal places).*

Question 15 (JAM 2025 • Q43 • ID: JAM_2025_Q43) NAT +1 M Neg: Nil (0)

Q.43 Let α be the real number such that

$$\lim_{x \to 0} \frac{(1 - \cos x)(2^{2+x} - 4)}{x^3} = \alpha \ln 2\ .$$

Then, the value of α is equal to _____ *(rounded off to two decimal places).*

Question 16 (JAM 2025 • Q42 • ID: JAM_2025_Q42) NAT +1 M Neg: Nil (0)

Q.42 The value of

$$\lim_{n \to \infty} 8n \left(e^{\left(\frac{1}{2n}\right)} - 1 \right) \left(\sin \frac{1}{2n} + \left| \cos \frac{1}{2n} \right| \right)$$

is equal to _____ *(rounded off to two decimal places).*

Question 17 (JAM 2025 • Q38 • ID: JAM_2025_Q38) MSQ +2 M Neg: Nil (0)

Q.38 Let $f: \mathbb{R} \to \mathbb{R}$ be a twice differentiable function such that

$$f(0) = 4, f(1) = -2, f(2) = 8 \text{ and } f(3) = 2\,.$$

Then, which of the following is/are TRUE?

(A) $|f'(x)| < 5$ for all $x \in [0,1]$

(B) $|f'(x_1)| \geq 5$ for some $x_1 \in [0,1]$

(C) $f'(x_2) = 0$ for some $x_2 \in [0,3]$

(D) $f''(x_3) = 0$ for some $x_3 \in [0,3]$

Question 18 (JAM 2025 • Q37 • ID: JAM_2025_Q37) MSQ +2 M Neg: Nil (0)

Q.37 Let $f: \mathbb{R} \to \mathbb{R}$ be a twice differentiable function such that

$$f(0) = 0, f'(0) = 2 \text{ and } f(1) = -3\,.$$

Then, which of the following is/are TRUE?

(A) $|f'(x)| \leq 2$ for all $x \in [0,1]$

(B) $|f'(x_1)| > 2$ for some $x_1 \in [0,1]$

(C) $|f''(x)| < 10$ for all $x \in [0,1]$

(D) $|f''(x_2)| \geq 10$ for some $x_2 \in [0,1]$

Question 19 (JAM 2025 • Q30 • ID: JAM_2025_Q30) MCQ +2 M Neg: −0.67

Q.30 The number of elements in the set

$$\{x \in \mathbb{R} : 8x^2 + x^4 + x^8 = \cos x\}$$

is equal to

(A) 0

(B) 1

(C) 2

(D) greater than or equal to 3

Question 20 (JAM 2025 • Q17 • ID: JAM_2025_Q17) MCQ +2 M Neg: −0.67

Q.17 Let $f(x) = 10x^2 + e^x - \sin(2x) - \cos x$, $x \in \mathbb{R}$. The number of points at which the function f has a local minimum is

(A) 0

(B) 1

(C) 2

(D) greater than or equal to 3

Question 21 (JAM 2025 • Q4 • ID: JAM_2025_Q4) MCQ +1 M Neg: −0.33

Q.4 Let $f, g : \mathbb{R} \to \mathbb{R}$ be two functions defined by

$$f(x) = \begin{cases} |x|^{1/8} \left|\sin\frac{1}{x}\right| \cos x & \text{if} \quad x \neq 0 \\ 0 & \text{if} \quad x = 0 \end{cases}$$

and

$$g(x) = \begin{cases} e^x \cos\frac{1}{x} & \text{if} \quad x \neq 0 \\ 1 & \text{if} \quad x = 0 \end{cases}.$$

Then, which one of the following is TRUE?

(A) f is continuous at $x = 0$, and g is NOT continuous at $x = 0$

(B) f is NOT continuous at $x = 0$, and g is continuous at $x = 0$

(C) f is continuous at $x = 0$, and g is continuous at $x = 0$

(D) f is NOT continuous at $x = 0$, and g is NOT continuous at $x = 0$

Question 22 (JAM 2025 • Q3 • ID: JAM_2025_Q3) MCQ +1 M Neg: −0.33

Q.3 Let $f, g: \mathbb{R} \to \mathbb{R}$ be two functions defined by

$$f(x) = \begin{cases} x\,|x|\left|\sin\frac{1}{x}\right| & \text{if} \quad x \neq 0 \\ 0 & \text{if} \quad x = 0 \end{cases}$$

and

$$g(x) = \begin{cases} x^2 \sin\frac{1}{x} + x\cos\frac{1}{x} & \text{if} \quad x \neq 0 \\ 0 & \text{if} \quad x = 0\,. \end{cases}$$

Then, which one of the following is TRUE?

(A) f is differentiable at $x = 0$, and g is NOT differentiable at $x = 0$

(B) f is NOT differentiable at $x = 0$, and g is differentiable at $x = 0$

(C) f is differentiable at $x = 0$, and g is differentiable at $x = 0$

(D) f is NOT differentiable at $x = 0$, and g is NOT differentiable at $x = 0$

Question 23 (JAM 2024 • Q60 • ID: JAM_2024_Q60) NAT +2 M Neg: Nil (0)

Q.60 Define the function $f: (-1,1) \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ by

$$f(x) = \sin^{-1} x$$

Let a_6 denote the coefficient of x^6 in the Taylor series of $\left(f(x)\right)^2$ about $x = 0$. Then, the value of $9a_6$ equals _____ (*rounded off to two decimal places*).

Question 24 (JAM 2024 • Q56 • ID: JAM_2024_Q56) NAT +2 M Neg: Nil (0)

Q.56 Consider the function $f:\mathbb{R}\to\mathbb{R}$ given by $f(x)=x^3-4x^2+4x-6$.

For $c\in\mathbb{R}$, let

$$S(c)=\left\{x\in\mathbb{R}: f(x)=c\right\}$$

and $|S(c)|$ denote the number of elements in $S(c)$. Then, the value of

$$|S(-7)|+|S(-5)|+|S(3)|$$

equals _____

Question 25 (JAM 2024 • Q52 • ID: JAM_2024_Q52) NAT +2 M Neg: Nil (0)

Q.52 The value of

$$\lim_{t\to\infty}\left(\left(\log\left(t^2+\frac{1}{t^2}\right)\right)^{-1}\int_1^{\pi t}\frac{\sin^2 5x}{x}\,dx\right)$$

equals _____ (*rounded off to two decimal places*).

Question 26 (JAM 2024 • Q46 • ID: JAM_2024_Q46) NAT +1 M Neg: Nil (0)

Q.46 Let

$$S = \left\{ f: \mathbb{R} \to \mathbb{R} : \ f \text{ is a polynomial and } f\big(f(x)\big) = \big(f(x)\big)^{2024} \ \text{ for } \ x \in \mathbb{R} \right\}.$$

Then, the number of elements in S is _____

Question 27 (JAM 2024 • Q38 • ID: JAM_2024_Q38) MSQ +2 M Neg: Nil (0)

Q.38 Define $f: \mathbb{R} \to \mathbb{R}$ and $g: \mathbb{R} \to \mathbb{R}$ as follows

$$f(x) = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{2^{2m}\ (m!)^2} \text{ and } g(x) = \frac{x}{2} \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{2^{2m}\ (m+1)!\ m!} \ \text{ for } \ x \in \mathbb{R}.$$

Let $x_1, x_2, x_3, x_4 \in \mathbb{R}$ be such that $0 < x_1 < x_2$, $0 < x_3 < x_4$,

$$f(x_1) = f(x_2) = 0, \qquad f(x) \neq 0 \text{ when } x_1 < x < x_2 ,$$

$$g(x_3) = g(x_4) = 0 \quad \text{and } g(x) \neq 0 \text{ when } x_3 < x < x_4.$$

Then, which of the following statements is/are TRUE?

(A) The function f does not vanish anywhere in the interval (x_3, x_4)

(B) The function f vanishes exactly once in the interval (x_3, x_4)

(C) The function g does not vanish anywhere in the interval (x_1, x_2)

(D) The function g vanishes exactly once in the interval (x_1, x_2)

Question 28 (JAM 2024 • Q34 • ID: JAM_2024_Q34) MSQ +2 M Neg: Nil (0)

Q.34 Let $f:(1,\infty)\to(0,\infty)$ be a continuous function such that for every $n\in\mathbb{N}$, $f(n)$ is the smallest prime factor of n. Then, which of the following options is/are CORRECT?

(A) $\lim_{x\to\infty} f(x)$ exists

(B) $\lim_{x\to\infty} f(x)$ does not exist

(C) The set of solutions to the equation $f(x)=2024$ is finite

(D) The set of solutions to the equation $f(x)=2024$ is infinite

Question 29 (JAM 2024 • Q29 • ID: JAM_2024_Q29) MCQ +2 M Neg: −0.67

Q.29 Let L_1 denote the line $y = 3x + 2$ and L_2 denote the line $y = 4x + 3$. Suppose that $f: \mathbb{R} \to \mathbb{R}$ is a four times continuously differentiable function such that the line L_1 intersects the curve $y = f(x)$ at exactly three distinct points and the line L_2 intersects the curve $y = f(x)$ at exactly four distinct points. Then, which one of the following is TRUE?

(A) $\frac{df}{dx}$ does not attain the value 3 on $\mathbb{R}$

(B) $\frac{d^2f}{dx^2}$ vanishes at most once on $\mathbb{R}$

(C) $\frac{d^3f}{dx^3}$ vanishes at least once on $\mathbb{R}$

(D) $\frac{df}{dx}$ does not attain the value $\frac{7}{2}$ on $\mathbb{R}$

Madras

Question 30 (JAM 2024 • Q25 • ID: JAM_2024_Q25) MCQ +2 M Neg: −0.67

Q.25 Let $f: \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \frac{(x^2+1)^2}{x^4+x^2+1} \quad \text{for } x \in \mathbb{R}.$$

Then, which one of the following is TRUE?

(A) f has exactly two points of local maxima and exactly three points of local minima

(B) f has exactly three points of local maxima and exactly two points of local minima

(C) f has exactly one point of local maximum and exactly two points of local minima

(D) f has exactly two points of local maxima and exactly one point of local minimum

Madras

Question 31 (JAM 2024 • Q24 • ID: JAM_2024_Q24) MCQ +2 M Neg: −0.67

Q.24 Consider the following two statements.

P: There exist functions $f: \mathbb{R} \to \mathbb{R}$, $g: \mathbb{R} \to \mathbb{R}$ such that f is continuous at $x = 1$ and g is discontinuous at $x = 1$ but $g \circ f$ is continuous at $x = 1$.

Q: There exist functions $f: \mathbb{R} \to \mathbb{R}$, $g: \mathbb{R} \to \mathbb{R}$ such that both f and g are discontinuous at $x = 1$ but $g \circ f$ is continuous at $x = 1$.

Which one of the following holds?

(A) Both P and Q are true

(B) Both P and Q are false

(C) P is true but Q is false

(D) P is false but Q is true

Madras

Question 32 (JAM 2024 • Q19 • ID: JAM_2024_Q19) MCQ +2 M Neg: −0.67

Q.19 For $x \in \mathbb{R}$, let $\lfloor x \rfloor$ denote the greatest integer less than or equal to x.

For $x, y \in \mathbb{R}$, define

$$\min\{x, y\} = \begin{cases} x & \text{if } x \leq y, \\ y & \text{otherwise.} \end{cases}$$

Let $f: [-2\pi, 2\pi] \to \mathbb{R}$ be defined by

$$f(x) = \sin(\min\{x, x - \lfloor x \rfloor\}) \text{ for } x \in [-2\pi, 2\pi].$$

Consider the set $S = \{x \in [-2\pi, 2\pi]: f \text{ is discontinuous at } x\}$.

Which one of the following statements is TRUE?

(A) S has 13 elements

(B) S has 7 elements

(C) S is an infinite set

(D) S has 6 elements

Question 33 (JAM 2023 • Q60 • ID: JAM_2023_Q60) NAT +2 M Neg: Nil (0)

Q. 60 Let $f : \mathbb{R} \to \mathbb{R}$ be a bijective function such that for all $x \in \mathbb{R}$, $f(x) = \sum_{n=1}^{\infty} a_n x^n$ and $f^{-1}(x) = \sum_{n=1}^{\infty} b_n x^n$, where f^{-1} is the inverse function of f. If $a_1 = 2$ and $a_2 = 4$, then b_1 is equal to _____.

Madras

Question 34 (JAM 2023 • Q52 • ID: JAM_2023_Q52) NAT +2 M Neg: Nil (0)

Q. 52 The global minimum value of

$$f(x) = |x-1| + |x-2|^2$$

on $\mathbb{R}$ is equal to _____. (rounded off to two decimal places)

Question 35 (JAM 2023 • Q47 • ID: JAM_2023_Q47) NAT +1 M Neg: Nil (0)

Q. 47 Let $f(x) = \sqrt[3]{x}$ for $x \in (0, \infty)$, and $\theta(h)$ be a function such that

$$f(3+h) - f(3) = hf'(3+\theta(h)h)$$

for all $h \in (-1, 1)$. Then $\lim_{h\to 0} \theta(h)$ is equal to _____.
(rounded off to two decimal places)

Question 36 (JAM 2023 • Q44 • ID: JAM_2023_Q44) NAT +1 M Neg: Nil (0)

Q. 44 The value of

$$\lim_{n\to\infty} \left(n \int_0^1 \frac{x^n}{x+1}\, dx \right)$$

is equal to _____. (rounded off to two decimal places)

Question 37 (JAM 2023 • Q35 • ID: JAM_2023_Q35) MSQ +2 M Neg: Nil (0)

Q. 35 Which of the following functions is/are Riemann integrable on $[0,1]$?

(A) $f(x) = \int_0^x \left|\frac{1}{2} - t\right| dt$

(B) $f(x) = \begin{cases} x\sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

(C) $f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \cap [0,1] \\ -1 & \text{otherwise} \end{cases}$

(D) $f(x) = \begin{cases} x & \text{if } x \in [0,1) \\ 0 & \text{if } x = 1 \end{cases}$

Question 38 (JAM 2023 • Q34 • ID: JAM_2023_Q34) MSQ +2 M Neg: Nil (0)

Q. 34 Let $f : (-1,1) \to \mathbb{R}$ be a differentiable function satisfying $f(0) = 0$. Suppose there exists an $M > 0$ such that $|f'(x)| \leq M|x|$ for all $x \in (-1,1)$. Then

(A) f' is continuous at $x = 0$

(B) f' is differentiable at $x = 0$

(C) ff' is differentiable at $x = 0$

(D) $(f')^2$ is differentiable at $x = 0$

Question 39 (JAM 2023 • Q30 • ID: JAM_2023_Q30) MCQ +2 M Neg: −0.67

Q. 30 Let $f : \mathbb{R} \to \mathbb{R}$ be an infinitely differentiable function such that f'' has exactly two distinct zeroes. Then

(A) f' has at most 3 distinct zeroes

(B) f' has at least 1 zero

(C) f has at most 3 distinct zeroes

(D) f has at least 2 distinct zeroes

Question 40 (JAM 2023 • Q28 • ID: JAM_2023_Q28) MCQ +2 M Neg: −0.67

Q. 28 Let $y : \mathbb{R} \to \mathbb{R}$ be a twice differentiable function such that y'' is continuous on $[0, 1]$ and $y(0) = y(1) = 0$. Suppose $y''(x) + x^2 < 0$ for all $x \in [0, 1]$. Then

(A) $y(x) > 0$ for all $x \in (0, 1)$

(B) $y(x) < 0$ for all $x \in (0, 1)$

(C) $y(x) = 0$ has exactly one solution in $(0, 1)$

(D) $y(x) = 0$ has more than one solution in $(0, 1)$

Question 41 (JAM 2023 • Q25 • ID: JAM_2023_Q25) MCQ +2 M Neg: −0.67

Q. 25 Let $f(x) = \cos(x)$ and $g(x) = 1 - \dfrac{x^2}{2}$ for $x \in \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$. Then

(A) $f(x) \geq g(x)$ for all $x \in \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$

(B) $f(x) \leq g(x)$ for all $x \in \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$

(C) $f(x) - g(x)$ changes sign exactly once on $\left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$

(D) $f(x) - g(x)$ changes sign more than once on $\left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$

Question 42 (JAM 2023 • Q6 • ID: JAM_2023_Q6) MCQ +1 M Neg: −0.33

Q. 6 The limit

$$\lim_{a \to 0} \left(\frac{\int_0^a \sin(x^2)\, dx}{\int_0^a (\ln(x+1))^2\, dx} \right)$$

is

(A) 0

(B) 1

(C) $\dfrac{\pi}{e}$

(D) non-existent

Question 43 (JAM 2023 • Q5 • ID: JAM_2023_Q5) MCQ +1 M Neg: −0.33

Q. 5 Let $p(x) = x^{57} + 3x^{10} - 21x^3 + x^2 + 21$ and

$$q(x) = p(x) + \sum_{j=1}^{57} p^{(j)}(x) \quad \text{for all } x \in \mathbb{R},$$

where $p^{(j)}(x)$ denotes the j^{th} derivative of $p(x)$. Then the function q admits

(A) NEITHER a global maximum NOR a global minimum on $\mathbb{R}$

(B) a global maximum but NOT a global minimum on $\mathbb{R}$

(C) a global minimum but NOT a global maximum on $\mathbb{R}$

(D) a global minimum and a global maximum on $\mathbb{R}$

Question 44 (JAM 2022 • Q60 • ID: JAM_2022_Q60) NAT +2 M Neg: Nil (0)

Q.60 If $f:[0,\infty) \to \mathbb{R}$ and $g:[0,\infty) \to [0,\infty)$ are continuous functions such that

$\int_0^{x^3+x^2} f(t)dt = x^2$ and $\int_0^{g(x)} t^2 dt = 9(x+1)^3$ for all $x \in [0,\infty)$,

then the value of

$$f(2) + g(2) + 16\, f(12)$$

is equal to _____. (*Rounded off to two decimal places*)

Question 45 (JAM 2022 • Q59 • ID: JAM_2022_Q59) NAT +2 M Neg: Nil (0)

Q.59 Let $f:(-1,1)\to\mathbb{R}$ and $g:(-1,1)\to\mathbb{R}$ be thrice continuously differentiable functions such that $f(x)\neq g(x)$ for every nonzero $x\in(-1,1)$. Suppose

$$f(0)=\ln 2,\quad f'(0)=\pi,\quad f''(0)=\pi^2,\quad \text{and}\quad f'''(0)=\pi^9$$

and

$$g(0)=\ln 2,\quad g'(0)=\pi,\quad g''(0)=\pi^2,\quad \text{and}\quad g'''(0)=\pi^3.$$

Then the value of the limit

$$\lim_{x\to 0}\frac{e^{f(x)}-e^{g(x)}}{f(x)-g(x)}$$

is equal to _____. (*Rounded off to two decimal places*)

Question 46 (JAM 2022 • Q54 • ID: JAM_2022_Q54) NAT +2 M Neg: Nil (0)

Q.54 Let $f:[0,\pi]\to\mathbb{R}$ be the function defined by

$$f(x)=\begin{cases}(x-\pi)e^{\sin x} & \text{if } 0\le x\le\dfrac{\pi}{2},\\ xe^{\sin x}+\dfrac{4}{\pi} & \text{if } \dfrac{\pi}{2}<x\le\pi.\end{cases}$$

Then the value of

$$\int_0^{\pi} f(x)dx$$

is equal to _____. (*Rounded off to two decimal places*)

Question 47 (JAM 2022 • Q41 • ID: JAM_2022_Q41) NAT +1 M Neg: Nil (0)

Q.41	The value of the limit $$\lim_{n\to\infty}\left(\frac{(1^4+2^4+\cdots+n^4)}{n^5}+\frac{1}{\sqrt{n}}\left(\frac{1}{\sqrt{n+1}}+\frac{1}{\sqrt{n+2}}+\cdots+\frac{1}{\sqrt{4n}}\right)\right)$$ is equal to _____. (*Rounded off to two decimal places*)

Question 48 (JAM 2022 • Q34 • ID: JAM_2022_Q34) MSQ +2 M Neg: Nil (0)

Q.34	Let S be the set of all functions $f:\mathbb{R}\to\mathbb{R}$ satisfying $$\lvert f(x)-f(y)\rvert^2 \le \lvert x-y\rvert^3 \quad \text{for all } x,y\in\mathbb{R}.$$ Then which of the following is/are true?
(A)	Every function in S is differentiable.
(B)	There exists a function $f\in S$ such that f is differentiable, but f is not twice differentiable.
(C)	There exists a function $f\in S$ such that f is twice differentiable, but f is not thrice differentiable.
(D)	Every function in S is infinitely differentiable.

Question 49 (JAM 2022 • Q33 • ID: JAM_2022_Q33) MSQ +2 M Neg: Nil (0)

Q.33	Define $f:[0,1] \to [0,1]$ by $$f(x) = \begin{cases} 1 & \text{if } x = 0, \\ \frac{1}{n} & \text{if } x = \frac{m}{n} \text{ for some } m, n \in \mathbb{N} \text{ with } m \leq n \text{ and } \gcd(m,n) = 1, \\ 0 & \text{if } x \in [0,1] \text{ is irrational.} \end{cases}$$ and define $g:[0,1] \to [0,1]$ by $$g(x) = \begin{cases} 0 & \text{if } x = 0, \\ 1 & \text{if } x \in (0,1]. \end{cases}$$ Then which of the following is/are true?
(A)	f is Riemann integrable on $[0,1]$.
(B)	g is Riemann integrable on $[0,1]$.
(C)	The composite function $f \circ g$ is Riemann integrable on $[0,1]$.
(D)	The composite function $g \circ f$ is Riemann integrable on $[0,1]$.

Question 50 (JAM 2022 • Q32 • ID: JAM_2022_Q32) MSQ +2 M Neg: Nil (0)

Q.32	Let S be the set of all continuous functions $f:[-1,1] \to \mathbb{R}$ satisfying the following three conditions: (i) f is infinitely differentiable on the open interval $(-1,1)$, (ii) the Taylor series $$f(0)+f'(0)x+\frac{f''(0)}{2!}x^2+\cdots$$ of f at 0 converges to $f(x)$ for each $x \in (-1,1)$, (iii) $f\left(\frac{1}{n}\right)=0$ for all $n \in \mathbb{N}$. Then which of the following is/are true?
(A)	$f(0)=0$ for every $f \in S$.
(B)	$f'\left(\frac{1}{2}\right)=0$ for every $f \in S$.
(C)	There exists $f \in S$ such that $f'\left(\frac{1}{2}\right) \neq 0$.
(D)	There exists $f \in S$ such that $f(x) \neq 0$ for some $x \in [-1,1]$.

Question 51 (JAM 2022 • Q29 • ID: JAM_2022_Q29) MCQ +2 M Neg: −0.67

Q.29	For $n \in \mathbb{N}$ and $x \in [1, \infty)$, let $$f_n(x) = \int_0^{\pi} \left(x^2 + (\cos\theta)\sqrt{x^2 - 1}\right)^n d\theta.$$ Then which one of the following is true?
(A)	$f_n(x)$ is not a polynomial in x if n is odd and $n \geq 3$.
(B)	$f_n(x)$ is not a polynomial in x if n is even and $n \geq 4$.
(C)	$f_n(x)$ is a polynomial in x for all $n \in \mathbb{N}$.
(D)	$f_n(x)$ is not a polynomial in x for any $n \geq 3$.

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Question 52 (JAM 2022 • Q27 • ID: JAM_2022_Q27) MCQ +2 M Neg: −0.67

Q.27	Let a, b be positive real numbers such that $a < b$. Given that $$\lim_{N\to\infty} \int_0^N e^{-t^2} dt = \frac{\sqrt{\pi}}{2},$$ the value of $$\lim_{N\to\infty} \int_0^N \frac{1}{t^2}\left(e^{-at^2} - e^{-bt^2}\right) dt$$ is equal to
(A)	$\sqrt{\pi}(\sqrt{a} - \sqrt{b})$.
(B)	$\sqrt{\pi}(\sqrt{a} + \sqrt{b})$.
(C)	$-\sqrt{\pi}(\sqrt{a} + \sqrt{b})$.
(D)	$\sqrt{\pi}(\sqrt{b} - \sqrt{a})$.

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Question 53 (JAM 2022 • Q22 • ID: JAM_2022_Q22) MCQ +2 M Neg: −0.67

Q.22	Let $H : \mathbb{R} \to \mathbb{R}$ be the function given by $H(x) = \frac{1}{2}(e^x + e^{-x})$ for $x \in \mathbb{R}$. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $$f(x) = \int_0^{\pi} H(x \sin\theta)\, d\theta \quad \text{for } x \in \mathbb{R}.$$ Then which one of the following is true?
(A)	$xf''(x) + f'(x) + xf(x) = 0$ for all $x \in \mathbb{R}$.
(B)	$xf''(x) - f'(x) + xf(x) = 0$ for all $x \in \mathbb{R}$.
(C)	$xf''(x) + f'(x) - xf(x) = 0$ for all $x \in \mathbb{R}$.
(D)	$xf''(x) - f'(x) - xf(x) = 0$ for all $x \in \mathbb{R}$.

Question 54 (JAM 2021 • Q57 • ID: JAM_2021_Q57) NAT +2 M Neg: Nil (0)

Q. 57 The value of

$$\lim_{n\to\infty} \int_0^1 e^{x^2} \sin(nx)\, dx$$

is _____.

Question 55 (JAM 2021 • Q54 • ID: JAM_2021_Q54) NAT +2 M Neg: Nil (0)

Q. 54 The largest positive number a such that

$$\int_0^5 f(x)dx + \int_0^3 f^{-1}(x)dx \geq a$$

for every strictly increasing surjective continuous function $f : [0, \infty) \to [0, \infty)$ is _____.

Question 56 (JAM 2021 • Q53 • ID: JAM_2021_Q53) NAT +2 M Neg: Nil (0)

Q. 53 Consider those continuous functions $f : \mathbb{R} \to \mathbb{R}$ that have the property that given any $x \in \mathbb{R}$,

$$f(x) \in \mathbb{Q} \text{ if and only if } f(x+1) \in \mathbb{R} \setminus \mathbb{Q}.$$

The number of such functions is _____.

Question 57 (JAM 2021 • Q45 • ID: JAM_2021_Q45) NAT +1 M Neg: Nil (0)

Q. 45 Consider the set $A = \{a \in \mathbb{R} : x^2 = a(a+1)(a+2) \text{ has a real root }\}$. The number of connected components of A is _____.

Question 58 (JAM 2021 • Q38 • ID: JAM_2021_Q38) MSQ +2 M Neg: Nil (0)

JAM 2021 MATHEMATICS - MA

Q. 38 Consider the four functions from $\mathbb{R}$ to $\mathbb{R}$:

$$f_1(x) = x^4 + 3x^3 + 7x + 1, \quad f_2(x) = x^3 + 3x^2 + 4x, \quad f_3(x) = \arctan(x)$$

and

$$f_4(x) = \begin{cases} x & \text{if } x \notin \mathbb{Z}, \\ 0 & \text{if } x \in \mathbb{Z}. \end{cases}$$

Which of the following subsets of $\mathbb{R}$ are open?

(A) The range of f_1.

(B) The range of f_2.

(C) The range of f_3.

(D) The range of f_4.

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Question 59 (JAM 2021 • Q35 • ID: JAM_2021_Q35) MSQ +2 M Neg: Nil (0)

Q. 35 Let $f : (a, b) \to \mathbb{R}$ be a differentiable function on (a, b). Which of the following statements is/are true?

(A) $f' > 0$ in (a, b) implies that f is increasing in (a, b).

(B) f is increasing in (a, b) implies that $f' > 0$ in (a, b).

(C) If $f'(x_0) > 0$ for some $x_0 \in (a, b)$, then there exists a $\delta > 0$ such that $f(x) > f(x_0)$ for all $x \in (x_0, x_0 + \delta)$.

(D) If $f'(x_0) > 0$ for some $x_0 \in (a, b)$, then f is increasing in a neighbourhood of x_0.

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Question 60 (JAM 2021 • Q32 • ID: JAM_2021_Q32) MSQ +2 M Neg: Nil (0)

Q. 32 Consider the equation

$$x^{2021} + x^{2020} + \cdots + x - 1 = 0.$$

Then

(A) all real roots are positive.

(B) exactly one real root is positive.

(C) exactly one real root is negative.

(D) no real root is positive.

2021

Question 61 (JAM 2021 • Q31 • ID: JAM_2021_Q31) MSQ +2 M Neg: Nil (0)

Q. 31 Let $f : \mathbb{R} \to \mathbb{R}$ be a function with the property that for every $y \in \mathbb{R}$, the value of the expression

$$\sup_{x \in \mathbb{R}} [xy - f(x)]$$

is finite. Define $g(y) = \sup_{x \in \mathbb{R}} [xy - f(x)]$ for $y \in \mathbb{R}$. Then

(A) g is even if f is even.

(B) f must satisfy $\lim_{|x| \to \infty} \frac{f(x)}{|x|} = +\infty$.

(C) g is odd if f is even.

(D) f must satisfy $\lim_{|x| \to \infty} \frac{f(x)}{|x|} = -\infty$.

Question 62 (JAM 2021 • Q26 • ID: JAM_2021_Q26) MCQ +2 M Neg: −0.67

Q. 26 Let $f : [0, 1] \to [0, 1]$ be a non-constant continuous function such that $f \circ f = f$. Define

$$E_f = \{x \in [0, 1] : f(x) = x\}.$$

Then

(A) E_f is neither open nor closed.

(B) E_f is an interval.

(C) E_f is empty.

(D) E_f need not be an interval.

Question 63 (JAM 2021 • Q21 • ID: JAM_2021_Q21) MCQ +2 M Neg: −0.67

Q. 21 Let $f : [0, 1] \to [0, \infty)$ be a continuous function such that

$$\big(f(t)\big)^2 < 1 + 2\int_0^t f(s)\, ds, \text{ for all } t \in [0, 1].$$

Then

(A) $f(t) < 1 + t$ for all $t \in [0, 1]$.

(B) $f(t) > 1 + t$ for all $t \in [0, 1]$.

(C) $f(t) = 1 + t$ for all $t \in [0, 1]$.

(D) $f(t) < 1 + \frac{t}{2}$ for all $t \in [0, 1]$.

Question 64 (JAM 2021 • Q18 • ID: JAM_2021_Q18) MCQ +2 M Neg: −0.67

JAM 2021 MATHEMATICS - MA

Q. 18 Consider the function

$$f(x) = \begin{cases} 1 & \text{if } x \in (\mathbb{R} \setminus \mathbb{Q}) \cup \{0\}, \\ 1 - \frac{1}{p} & \text{if } x = \frac{n}{p},\ n \in \mathbb{Z} \setminus \{0\},\ p \in \mathbb{N} \text{ and } \gcd(n,p) = 1. \end{cases}$$

Then

(A) all $x \in \mathbb{Q} \setminus \{0\}$ are strict local minima for f.

(B) f is continuous at all $x \in \mathbb{Q}$.

(C) f is not continuous at all $x \in \mathbb{R} \setminus \mathbb{Q}$.

(D) f is not continuous at $x = 0$.

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Question 65 (JAM 2021 • Q17 • ID: JAM_2021_Q17) MCQ +2 M Neg: −0.67

Q. 17 Let $f : \mathbb{R} \to \mathbb{R}$ be an infinitely differentiable function such that for all $a, b \in \mathbb{R}$ with $a < b$,

$$\frac{f(b) - f(a)}{b - a} = f'\Big(\frac{a+b}{2}\Big).$$

Then

(A) f must be a polynomial of degree less than or equal to 2.

(B) f must be a polynomial of degree greater than 2.

(C) f is not a polynomial.

(D) f must be a linear polynomial.

JAM 20

Question 66 (JAM 2021 • Q16 • ID: JAM_2021_Q16) MCQ +2 M Neg: −0.67

Q. 16 Define

$$S = \lim_{n\to\infty}\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)\cdots\left(1-\frac{1}{n^2}\right).$$

Then

(A) $S = 1/2$. (B) $S = 1/4$. (C) $S = 1$. (D) $S = 3/4$.

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Question 67 (JAM 2021 • Q7 • ID: JAM_2021_Q7) MCQ +1 M Neg: −0.33

JAM 2021 MATHEMATICS - MA

Q. 7 For every $n \in \mathbb{N}$, let $f_n : \mathbb{R} \to \mathbb{R}$ be a function. From the given choices, pick the statement that is the negation of

"For every $x \in \mathbb{R}$ and for every real number $\epsilon > 0$, there exists an integer $N > 0$ such that $\sum_{i=1}^{p} |f_{N+i}(x)| < \epsilon$ for every integer $p > 0$."

(A) For every $x \in \mathbb{R}$ and for every real number $\epsilon > 0$, there does not exist any integer $N > 0$ such that $\sum_{i=1}^{p} |f_{N+i}(x)| < \epsilon$ for every integer $p > 0$.

(B) For every $x \in \mathbb{R}$ and for every real number $\epsilon > 0$, there exists an integer $N > 0$ such that $\sum_{i=1}^{p} |f_{N+i}(x)| \geq \epsilon$ for some integer $p > 0$.

(C) There exists $x \in \mathbb{R}$ and there exists a real number $\epsilon > 0$ such that for every integer $N > 0$, there exists an integer $p > 0$ for which the inequality $\sum_{i=1}^{p} |f_{N+i}(x)| \geq \epsilon$ holds.

(D) There exists $x \in \mathbb{R}$ and there exists a real number $\epsilon > 0$ such that for every integer $N > 0$ and for every integer $p > 0$ the inequality $\sum_{i=1}^{p} |f_{N+i}(x)| \geq \epsilon$ holds.

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Question 68 (JAM 2021 • Q4 • ID: JAM_2021_Q4) MCQ +1 M Neg: −0.33

JAM 2021 MATHEMATICS - MA

Q. 4 Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function such that for all $x \in \mathbb{R}$,

$$\int_0^1 f(xt)\,dt = 0. \qquad (*)$$

Then

(A) f must be identically 0 on the whole of $\mathbb{R}$.

(B) there is an f satisfying $(*)$ that is identically 0 on $(0, 1)$ but not identically 0 on the whole of $\mathbb{R}$.

(C) there is an f satisfying $(*)$ that takes both positive and negative values.

(D) there is an f satisfying $(*)$ that is 0 at infinitely many points, but is not identically zero.

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Question 69 (JAM 2021 • Q3 • ID: JAM_2021_Q3) MCQ +1 M Neg: −0.33

Q. 3 Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function satisfying $f(x) = f(x + 1)$ for all $x \in \mathbb{R}$. Then

(A) f is not necessarily bounded above.

(B) there exists a unique $x_0 \in \mathbb{R}$ such that $f(x_0 + \pi) = f(x_0)$.

(C) there is no $x_0 \in \mathbb{R}$ such that $f(x_0 + \pi) = f(x_0)$.

(D) there exist infinitely many $x_0 \in \mathbb{R}$ such that $f(x_0 + \pi) = f(x_0)$.

JAM

MA 1 / 17

Question 70 (JAM 2020 • Q54 • ID: JAM_2020_Q54) NAT +2 M Neg: Nil (0)

Q. 54 Let $f(x) = \sqrt{x} + \alpha x,\ x > 0$ and

$$g(x) = a_0 + a_1(x-1) + a_2(x-1)^2$$

be the sum of the first three terms of the Taylor series of $f(x)$ around $x = 1$. If $g(3) = 3$, then α is _____.

Question 71 (JAM 2020 • Q44 • ID: JAM_2020_Q44) NAT +1 M Neg: Nil (0)

Q. 44 Let $S = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$ and $f : S \to \mathbb{R}$ be defined by $f(x) = \frac{1}{x}$. Then

$$\max\left\{\delta : \left|x - \frac{1}{3}\right| < \delta \Longrightarrow \left|f(x) - f\left(\frac{1}{3}\right)\right| < 1\right\}$$

is _____. (rounded off to two decimal places)

Question 72 (JAM 2020 • Q43 • ID: JAM_2020_Q43) NAT +1 M Neg: Nil (0)

Q. 43 Let $f : \mathbb{R} \to \mathbb{R}$ be such that f, f', f'' are continuous functions with $f > 0, f' > 0$ and $f'' > 0$. Then

$$\lim_{x \to -\infty} \frac{f(x) + f'(x)}{2}$$

is _____.

Question 73 (JAM 2020 • Q33 • ID: JAM_2020_Q33) MSQ +2 M Neg: Nil (0)

Q. 33 Let $a, b \in \mathbb{R}$ and $a < b$. Which of the following statement(s) is/are true?

(A) There exists a continuous function $f : [a, b] \to (a, b)$ such that f is one-one

(B) There exists a continuous function $f : [a, b] \to (a, b)$ such that f is onto

(C) There exists a continuous function $f : (a, b) \to [a, b]$ such that f is one-one

(D) There exists a continuous function $f : (a, b) \to [a, b]$ such that f is onto

Question 74 (JAM 2020 • Q31 • ID: JAM_2020_Q31) MSQ +2 M Neg: Nil (0)

Q. 31 Let $a, b, c \in \mathbb{R}$ such that $a < b < c$. Which of the following is/are true for any continuous function $f : \mathbb{R} \to \mathbb{R}$ satisfying $f(a) = b, f(b) = c$ and $f(c) = a$?

(A) There exists $\alpha \in (a, c)$ such that $f(\alpha) = \alpha$

(B) There exists $\beta \in (a, b)$ such that $f(\beta) = \beta$

(C) There exists $\gamma \in (a, b)$ such that $(f \circ f)(\gamma) = \gamma$

(D) There exists $\delta \in (a, c)$ such that $(f \circ f \circ f)(\delta) = \delta$

Question 75 (JAM 2020 • Q28 • ID: JAM_2020_Q28) MCQ +2 M Neg: −0.67

Q. 28 Let $f : [0,1] \to \mathbb{R}$ be a continuous function such that $f\left(\frac{1}{2}\right) = -\frac{1}{2}$ and

$$|f(x) - f(y) - (x-y)| \le \sin\left(|x-y|^2\right)$$

for all $x, y \in [0,1]$. Then $\int_0^1 f(x)\,dx$ is

(A) $-\frac{1}{2}$ (B) $-\frac{1}{4}$ (C) $\frac{1}{4}$ (D) $\frac{1}{2}$

Question 76 (JAM 2020 • Q14 • ID: JAM_2020_Q14) MCQ +2 M Neg: −0.67

Q. 14 Let $a \in \mathbb{R}$. If $f(x) = \begin{cases} (x+a)^2, & x \le 0 \\ (x+a)^3, & x > 0, \end{cases}$

then

(A) $\frac{d^2 f}{dx^2}$ does not exist at $x = 0$ for any value of a

(B) $\frac{d^2 f}{dx^2}$ exists at $x = 0$ for exactly one value of a

(C) $\frac{d^2 f}{dx^2}$ exists at $x = 0$ for exactly two values of a

(D) $\frac{d^2 f}{dx^2}$ exists at $x = 0$ for infinitely many values of a

$$\begin{cases} x^2 \sin\frac{1}{x} + y^2 \sin\frac{1}{y}, & xy \neq 0 \end{cases}$$

Question 77 (JAM 2020 • Q3 • ID: JAM_2020_Q3) MCQ +1 M Neg: −0.33

Q. 3 Which of the following is FALSE?

(A) $\lim_{x\to\infty} \frac{x}{e^x} = 0$ (B) $\lim_{x\to 0^+} \frac{1}{xe^{1/x}} = 0$

(C) $\lim_{x\to 0^+} \frac{\sin x}{1+2x} = 0$ (D) $\lim_{x\to 0^+} \frac{\cos x}{1+2x} = 0$

Question 78 (JAM 2020 • Q2 • ID: JAM_2020_Q2) MCQ +1 M Neg: −0.33

Q. 2 Let $f(x) = 2x^3 - 9x^2 + 7$. Which of the following is true?

(A) f is one-one in the interval $[-1, 1]$

(B) f is one-one in the interval $[2, 4]$

(C) f is NOT one-one in the interval $[-4, 0]$

(D) f is NOT one-one in the interval $[0, 4]$

Question 79 (JAM 2019 • Q47 • ID: JAM_2019_Q47) NAT +1 M Neg: Nil (0)

Q.47 If

$$g(x) = \int_{x(x-2)}^{4x-5} f(t)\,dt\,, \text{ where } f(x) = \sqrt{1+3x^4} \text{ for } x \in \mathbb{R}$$

then $g'(1) =$ _____

Question 80 (JAM 2019 • Q46 • ID: JAM_2019_Q46) NAT +1 M Neg: Nil (0)

Q.46 Let $f: [0, 1] \to \mathbb{R}$ be given by

$$f(x) = \frac{\left(1+x^{\frac{1}{3}}\right)^3 + \left(1-x^{\frac{1}{3}}\right)^3}{8(1+x)}.$$

Then

$$\max\{f(x): x \in [0,1]\} - \min\{f(x): x \in [0,1]\}$$

is _____

Question 81 (JAM 2019 • Q45 • ID: JAM_2019_Q45) NAT +1 M Neg: Nil (0)

Q.45 The coefficient of $\left(x-\frac{\pi}{2}\right)$ in the Taylor series expansion of the function

$$f(x)=\begin{cases}\dfrac{4(1-\sin x)}{2x-\pi}, & x\neq\dfrac{\pi}{2}\\ 0, & x=\dfrac{\pi}{2}\end{cases}$$

about $x=\frac{\pi}{2}$, is _____

Question 82 (JAM 2019 • Q38 • ID: JAM_2019_Q38) MSQ +2 M Neg: Nil (0)

Q.38 Let $f:\left(0,\frac{\pi}{2}\right)\to\mathbb{R}$ be given by

$$f(x)=(\sin x)^{\pi}-\pi\sin x+\pi.$$

Then which of the following statements is/are TRUE?

(A) f is an increasing function
(B) f is a decreasing function
(C) $f(x)>0$ for all $x\in\left(0,\frac{\pi}{2}\right)$
(D) $f(x)<0$ for some $x\in\left(0,\frac{\pi}{2}\right)$

Question 83 (JAM 2019 • Q37 • ID: JAM_2019_Q37) MSQ +2 M Neg: Nil (0)

Q.37 Let

$$f(x) = \cos(|\pi - x|) + (x - \pi)\sin|x| \text{ and } g(x) = x^2 \text{ for } x \in \mathbb{R}.$$

If $h(x) = f\big(g(x)\big)$, then

(A) h is not differentiable at $x = 0$
(B) $h'\big(\sqrt{\pi}\big) = 0$
(C) $h''(x) = 0$ has a solution in $(-\pi, \pi)$
(D) there exists $x_0 \in (-\pi, \pi)$ such that $h(x_0) = x_0$

Question 84 (JAM 2019 • Q18 • ID: JAM_2019_Q18) MCQ +2 M Neg: −0.67

Q.18 For which one of the following values of k, the equation

$$2x^3 + 3x^2 - 12x - k = 0$$

has three distinct real roots?

(A) 16 (B) 20 (C) 26 (D) 31

Question 85 (JAM 2019 • Q17 • ID: JAM_2019_Q17) MCQ +2 M Neg: −0.67

Q.17 Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function such that $f'(x) > f(x)$ for all $x \in \mathbb{R}$, and $f(0) = 1$. Then $f(1)$ lies in the interval

(A) $(0,\ e^{-1})$ (B) $\big(e^{-1}, \sqrt{e}\big)$ (C) $(\sqrt{e},\ e)$ (D) (e, ∞)

Question 86 (JAM 2019 • Q16 • ID: JAM_2019_Q16) MCQ +2 M Neg: −0.67

Q.16 Let $f(x) = (\ln x)^2$, $x > 0$. Then

(A) $\lim\limits_{x\to\infty} \frac{f(x)}{x}$ does not exist
(B) $\lim\limits_{x\to\infty} f'(x) = 2$
(C) $\lim\limits_{x\to\infty} \big(f(x+1) - f(x)\big) = 0$
(D) $\lim\limits_{x\to\infty} \big(f(x+1) - f(x)\big)$ does not exist

Question 87 (JAM 2018 • Q55 • ID: JAM_2018_Q55) NAT +2 M Neg: Nil (0)

Q.55 If $\alpha = \int_{\pi/6}^{\pi/3} \frac{\sin t + \cos t}{\sqrt{\sin 2t}}\, dt$, then the value of $\left(2\sin\frac{\alpha}{2} + 1\right)^2$ is _____

Question 88 (JAM 2018 • Q52 • ID: JAM_2018_Q52) NAT +2 M Neg: Nil (0)

Q.52 Let $f: \mathbb{R} \to \mathbb{R}$ be such that f'' is continuous on $\mathbb{R}$ and $f(0) = 1$, $f'(0) = 0$ and $f''(0) = -1$. Then $\lim_{x\to\infty} \left(f\left(\sqrt{\frac{2}{x}}\right)\right)^x$ is _____ (correct up to three decimal places).

Question 89 (JAM 2018 • Q50 • ID: JAM_2018_Q50) NAT +1 M Neg: Nil (0)

Q.50 The coefficient of x^4 in the power series expansion of $e^{\sin x}$ about $x = 0$ is _____ (correct up to three decimal places).

Q. 51 – Q. 60 carry two marks each.

Question 90 (JAM 2018 • Q31 • ID: JAM_2018_Q31) MSQ +2 M Neg: Nil (0)

Q. 31 – Q. 40 carry two marks each.

Q.31 Let $f: \mathbb{R}\setminus\{0\} \to \mathbb{R}$ be defined by $f(x) = x + \frac{1}{x^3}$. On which of the following interval(s) is f one-one?

(A) $(-\infty, -1)$ (B) $(0, 1)$ (C) $(0, 2)$ (D) $(0, \infty)$

Question 91 (JAM 2018 • Q30 • ID: JAM_2018_Q30) MCQ +2 M Neg: −0.67

Q.30 For $x > \frac{-1}{2}$, let $f_1(x) = \frac{2x}{1+2x}$, $f_2(x) = \log_e(1+2x)$ and $f_3(x) = 2x$. Then which one of the following is TRUE?

(A) $f_3(x) < f_2(x) < f_1(x)$ for $0 < x < \frac{\sqrt{3}}{2}$

(B) $f_1(x) < f_3(x) < f_2(x)$ for $x > 0$

(C) $f_1(x) + f_2(x) < \frac{f_3(x)}{2}$ for $x > \frac{\sqrt{3}}{2}$

(D) $f_2(x) < f_1(x) < f_3(x)$ for $x > 0$

SECTION - B

MULTIPLE SELECT QUESTIONS (MSQ)

Q. 31 – Q. 40 carry two marks each.

Question 92 (JAM 2018 • Q29 • ID: JAM_2018_Q29) MCQ +2 M Neg: −0.67

Q.29 Let $f: \mathbb{R} \to \mathbb{R}$ be a function and let J be a bounded open interval in $\mathbb{R}$. Define

$$W(f, J) = \sup\{f(x) \mid x \in J\} - \inf\{f(x) \mid x \in J\}.$$

Which one of the following is FALSE?

(A) $W(f, J_1) \leq W(f, J_2)$ if $J_1 \subset J_2$

(B) If f is a bounded function in J and $J \supset J_1 \supset J_2 \cdots \supset J_n \supset \cdots$ such that the length of the interval J_n tends to 0 as $n \to \infty$, then $\lim_{n\to\infty} W(f, J_n) = 0$

(C) If f is discontinuous at a point $a \in J$, then $W(f, J) \neq 0$

(D) If f is continuous at a point $a \in J$, then for any given $\epsilon > 0$ there exists an interval $I \subset J$ such that $W(f, I) < \epsilon$

Question 93 (JAM 2018 • Q16 • ID: JAM_2018_Q16) MCQ +2 M Neg: −0.67

Q.16 For $x \in \mathbb{R}$, let $f(x) = \begin{cases} x^3 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$. Then which one of the following is FALSE?

(A) $\lim_{x\to 0} \frac{f(x)}{x} = 0$

(B) $\lim_{x\to 0} \frac{f(x)}{x^2} = 0$

(C) $\frac{f(x)}{x^2}$ has infinitely many maxima and minima on the interval $(0,1)$

(D) $\frac{f(x)}{x^4}$ is continuous at $x = 0$ but not differentiable at $x = 0$

Question 94 (JAM 2018 • Q15 • ID: JAM_2018_Q15) MCQ +2 M Neg: −0.67

Q.15 Suppose that $f, g : \mathbb{R} \to \mathbb{R}$ are differentiable functions such that f is strictly increasing and g is strictly decreasing. Define $p(x) = f(g(x))$ and $q(x) = g(f(x))$, $\forall x \in \mathbb{R}$. Then, for $t > 0$, the sign of $\int_0^t p'(x)\,(q'(x) - 3)dx$ is

(A) positive (B) negative (C) dependent on t (D) dependent on f and g

Question 95 (JAM 2018 • Q4 • ID: JAM_2018_Q4) MCQ +1 M Neg: −0.33

Q.4 Let a be a positive real number. If f is a continuous and even function defined on the interval $[-a, a]$, then $\int_{-a}^{a} \frac{f(x)}{1+e^x}\,dx$ is equal to

(A) $\int_0^a f(x)\,dx$ (B) $2\int_0^a \frac{f(x)}{1+e^x}\,dx$

(C) $2\int_0^a f(x)\,dx$ (D) $2a\int_0^a \frac{f(x)}{1+e^x}\,dx$

Question 96 (JAM 2017 • Q57 • ID: JAM_2017_Q57) NAT +2 M Neg: Nil (0)

Q.57 For $x > 1$, let

$$f(x) = \int_1^x \left(\sqrt{\log t} - \frac{1}{2}\log\sqrt{t}\right) dt$$

The number of tangents to the curve $y = f(x)$ parallel to the line $x + y = 0$ is _____

Question 97 (JAM 2017 • Q53 • ID: JAM_2017_Q53) NAT +2 M Neg: Nil (0)

Q.53 Let $f(x) = \frac{\sin \pi x}{\pi \sin x}$, $x \in (0, \pi)$, and let $x_0 \in (0, \pi)$ be such that $f'(x_0) = 0$. Then

$$\left(f(x_0)\right)^2 (1 + (\pi^2 - 1)\sin^2 x_0) = _____$$

Question 98 (JAM 2017 • Q49 • ID: JAM_2017_Q49) NAT +1 M Neg: Nil (0)

Q.49 For $x > 0$, let $\lfloor x \rfloor$ denote the greatest integer less than or equal to x. Then

$$\lim_{x \to 0^+} x\left(\left\lfloor \frac{1}{x} \right\rfloor + \left\lfloor \frac{2}{x} \right\rfloor + \cdots + \left\lfloor \frac{10}{x} \right\rfloor\right) = _____$$

Question 99 (JAM 2017 • Q45 • ID: JAM_2017_Q45) NAT +1 M Neg: Nil (0)

Q.45

$$\left(\int_0^1 x^4 (1 - x)^5 dx\right)^{-1} = _____$$

Question 100 (JAM 2017 • Q25 • ID: JAM_2017_Q25) MCQ +2 M Neg: −0.67

Q.25 $\sum_{n=1}^{\infty} \tan^{-1}\frac{2}{n^2} =$

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{4}$ (D) π

Question 101 (JAM 2017 • Q20 • ID: JAM_2017_Q20) MCQ +2 M Neg: −0.67

Q.20 Let $f:\mathbb{R}\to[0,\infty)$ be a continuous function. Then which one of the following is NOT TRUE?

(A) There exists $x\in\mathbb{R}$ such that $f(x)=\frac{f(0)+f(1)}{2}$

(B) There exists $x\in\mathbb{R}$ such that $f(x)=\sqrt{f(-1)f(1)}$

(C) There exists $x\in\mathbb{R}$ such that $f(x)=\int_{-1}^{1} f(t)dt$

(D) There exists $x\in\mathbb{R}$ such that $f(x)=\int_{0}^{1} f(t)dt$

Question 102 (JAM 2017 • Q19 • ID: JAM_2017_Q19) MCQ +2 M Neg: −0.67

Q.19 Let $f:\mathbb{R}\to\mathbb{R}$ be a differentiable function such that $f(2)=2$ and

$$|f(x)-f(y)|\le 5(|x-y|)^{3/2}$$

for all $x\in\mathbb{R}, y\in\mathbb{R}$. Let $g(x)=x^3f(x)$. Then $g'(2)=$

(A) 5 (B) $\frac{15}{2}$ (C) 12 (D) 24

Question 103 (JAM 2017 • Q10 • ID: JAM_2017_Q10) MCQ +1 M Neg: −0.33

Q.10 If

$$f(x) = \begin{cases} 1+x & \text{if } x < 0 \\ (1-x)(px+q) & \text{if } x \geq 0 \end{cases}$$

satisfies the assumptions of Rolle's theorem in the interval $[-1, 1]$, then the ordered pair (p, q) is

(A) $(2, -1)$ (B) $(-2, -1)$ (C) $(-2, 1)$ (D) $(2, 1)$

Question 104 (JAM 2017 • Q8 • ID: JAM_2017_Q8) MCQ +1 M Neg: −0.33

Q.8 Let

$$f(x) = \frac{x + |x|(1+x)}{x} \sin\left(\frac{1}{x}\right), \qquad x \neq 0.$$

Write $L = \lim_{x \to 0^-} f(x)$ and $R = \lim_{x \to 0^+} f(x)$. Then which one of the following is TRUE?

(A) L exists but R does not exist

(B) L does not exist but R exists

(C) Both L and R exist

(D) Neither L nor R exists

Question 105 (JAM 2017 • Q7 • ID: JAM_2017_Q7) MCQ +1 M Neg: −0.33

Q.7 Let $f_1(x), f_2(x), g_1(x), g_2(x)$ be differentiable functions on $\mathbb{R}$. Let $F(x) = \begin{vmatrix} f_1(x) & f_2(x) \\ g_1(x) & g_2(x) \end{vmatrix}$ be the determinant of the matrix $\begin{bmatrix} f_1(x) & f_2(x) \\ g_1(x) & g_2(x) \end{bmatrix}$. Then $F'(x)$ is equal to

(A) $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1(x) & g_2(x) \end{vmatrix} + \begin{vmatrix} f_1(x) & g_1'(x) \\ f_2'(x) & g_2(x) \end{vmatrix}$

(B) $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1(x) & g_2(x) \end{vmatrix} + \begin{vmatrix} f_1(x) & g_1'(x) \\ f_2(x) & g_2'(x) \end{vmatrix}$

(C) $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1(x) & g_2(x) \end{vmatrix} - \begin{vmatrix} f_1(x) & g_1'(x) \\ f_2(x) & g_2'(x) \end{vmatrix}$

(D) $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1'(x) & g_2'(x) \end{vmatrix}$

Question 106 (JAM 2017 • Q2 • ID: JAM_2017_Q2) MCQ +1 M Neg: −0.33

Q.2 Let $\varphi: \mathbb{R} \to \mathbb{R}$ be a differentiable function such that φ' is strictly increasing with $\varphi'(1) = 0$. Let α and β denote the minimum and maximum values of $\varphi(x)$ on the interval $[2, 3]$, respectively. Then which one of the following is TRUE?

(A) $\beta = \varphi(3)$ (B) $\alpha = \varphi(2.5)$ (C) $\beta = \varphi(2.5)$ (D) $\alpha = \varphi(3)$

Question 107 (JAM 2016 • Q49 • ID: JAM_2016_Q49) NAT +1 M Neg: Nil (0)

Q.49 Let $f: (0, \infty) \to \mathbb{R}$ be a continuous function such that

$$\int_0^x f(t)dt = -2 + \frac{x^2}{2} + 4x \sin 2x + 2\cos 2x.$$

Then the value of $\frac{1}{\pi} f\left(\frac{\pi}{4}\right)$ is _____

Question 108 (JAM 2016 • Q46 • ID: JAM_2016_Q46) NAT +1 M Neg: Nil (0)

Q.46 If $f: (-1, \infty) \to \mathbb{R}$ defined by $f(x) = \frac{x}{1+x}$ is expressed as

$$f(x) = \frac{2}{3} + \frac{1}{9}(x-2) + \frac{c(x-2)^2}{(1+\xi)^3},$$

where ξ lies between 2 and x, then the value of c is _____

Question 109 (JAM 2016 • Q37 • ID: JAM_2016_Q37) MSQ +2 M Neg: Nil (0)

Q.37 Let $P(x) = \left(\frac{5}{13}\right)^x + \left(\frac{12}{13}\right)^x - 1$ for all $x \in \mathbb{R}$. Then which of the following statement(s) is(are) TRUE?

(A) The equation $P(x) = 0$ has exactly one solution in $\mathbb{R}$
(B) $P(x)$ is strictly increasing for all $x \in \mathbb{R}$
(C) The equation $P(x) = 0$ has exactly two solutions in $\mathbb{R}$
(D) $P(x)$ is strictly decreasing for all $x \in \mathbb{R}$

Question 110 (JAM 2016 • Q32 • ID: JAM_2016_Q32) MSQ +2 M Neg: Nil (0)

Q.32 The value(s) of the integral

$$\int_{-\pi}^{\pi} |x| \cos nx \, dx \ , \ n \geq 1$$

is (are)

(A) 0 when n is even
(B) 0 when n is odd
(C) $-\frac{4}{n^2}$ when n is even
(D) $-\frac{4}{n^2}$ when n is odd

Question 111 (JAM 2016 • Q30 • ID: JAM_2016_Q30) MCQ +2 M Neg: −0.67

Q.30 Let $f: \mathbb{R} \to \mathbb{R}$ be defined as

$$f(x) = \begin{cases} x(1 + x^{\alpha} \sin(\ln x^2)) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}.$$

Then, at $x = 0$, the function f is

(A) continuous and differentiable when $\alpha = 0$
(B) continuous and differentiable when $\alpha > 0$
(C) continuous and differentiable when $-1 < \alpha < 0$
(D) continuous and differentiable when $\alpha < -1$

SECTION - B

MULTIPLE SELECT QUESTIONS (MSQ)

Q. 31 – Q. 40 carry two marks each.

Question 112 (JAM 2016 • Q25 • ID: JAM_2016_Q25) MCQ +2 M Neg: −0.67

Q.25 Let $f(x) = \frac{1}{1+|x|} + \frac{1}{1+|x-1|}$ for all $x \in [-1, 1]$. Then which one of the following is TRUE?

(A) Maximum value of $f(x)$ is $\frac{3}{2}$
(B) Minimum value of $f(x)$ is $\frac{1}{3}$
(C) Maximum of $f(x)$ occurs at $x = \frac{1}{2}$
(D) Minimum of $f(x)$ occurs at $x = 1$

Question 113 (JAM 2016 • Q14 • ID: JAM_2016_Q14) MCQ +2 M Neg: −0.67

Q.14 The value of the integral

$$\frac{(2n)!}{2^{2n}\,(n!)} \int_{-1}^{1} (1 - x^2)^n \, dx, \qquad n \in \mathbb{N}$$

is

(A) $\frac{2}{(2n+1)!}$ (B) $\frac{2n}{(2n+1)!}$

(C) $\frac{2\,(n!)}{2n+1}$ (D) $\frac{(n+1)!}{2n+1}$

Question 114 (JAM 2016 • Q6 • ID: JAM_2016_Q6) MCQ +1 M Neg: −0.33

Q.6 The value of $\lim\limits_{(x,\ y)\to(2,-2)} \frac{\sqrt{(x-y)}-2}{x-y-4}$ is

(A) 0 (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

Question 115 (JAM 2016 • Q5 • ID: JAM_2016_Q5) MCQ +1 M Neg: −0.33

Q.5 Let f be a strictly monotonic continuous real valued function defined on $[a, b]$ such that $f(a) < a$ and $f(b) > b$. Then which one of the following is TRUE?

(A) There exists exactly one $c \in (a, b)$ such that $f(c) = c$
(B) There exist exactly two points $c_1, c_2 \in (a, b)$ such that $f(c_i) = c_i,\ i = 1,2$
(C) There exists no $c \in (a, b)$ such that $f(c) = c$
(D) There exist infinitely many points $c \in (a, b)$ such that $f(c) = c$

Question 116 (JAM 2016 • Q3 • ID: JAM_2016_Q3) MCQ +1 M Neg: −0.33

Q.3 Let $f: [-1, 1] \to \mathbb{R}$ be a continuous function. Then the integral

$$\int_0^{\pi} x\, f(\sin x)\ dx$$

is equivalent to

(A) $\frac{\pi}{2}\int_0^{\pi} f(\sin x)\ dx$

(B) $\frac{\pi}{2}\int_0^{\pi} f(\cos x)\ dx$

(C) $\pi\int_0^{\pi} f(\cos\ x)\ dx$

(D) $\pi\int_0^{\pi} f(\sin x)\ dx$

Question 117 (JAM 2015 • Q59 • ID: JAM_2015_Q59) NAT +2 M Neg: Nil (0)

Q.19 The limit

$$\lim_{x\to 0+} \frac{9}{x}\left(\frac{1}{\tan^{-1} x} - \frac{1}{x}\right)$$

is equal to _____

Question 118 (JAM 2015 • Q56 • ID: JAM_2015_Q56) NAT +2 M Neg: Nil (0)

Q.16

The coefficient of $\left(x-\frac{\pi}{4}\right)^3$ in the Taylor series expansion of the function

$$f(x)=3\sin x\cos\left(x+\frac{\pi}{4}\right),\qquad x\in\mathbb{R}$$

about the point $\frac{\pi}{4}$, correct upto three decimal places, is _____

Question 119 (JAM 2015 • Q53 • ID: JAM_2015_Q53) NAT +2 M Neg: Nil (0)

Q.13 The limit

$$\lim_{n\to\infty}\sum_{k=2}^{n}\frac{1}{k^3-k}$$

is equal to _____

Question 120 (JAM 2015 • Q50 • ID: JAM_2015_Q50) NAT +1 M Neg: Nil (0)

Q.10

Let $f:(0,1)\to\mathbb{R}$ be a continuously differentiable function such that f' has finitely many zeros in $(0,1)$ and f' changes sign at exactly two of these points. Then for any $y\in\mathbb{R}$, the maximum number of solutions to $f(x)=y$ in $(0,1)$ is _____

Q. 11 – Q. 20 carry two marks each

Question 121 (JAM 2015 • Q49 • ID: JAM_2015_Q49) NAT +1 M Neg: Nil (0)

Q.9

Let $f:\mathbb{R}\longrightarrow\mathbb{R}$ be defined by

$$f(x)=\begin{cases}x^6-1, & x\in\mathbb{Q}\\ 1-x^6, & x\notin\mathbb{Q}\end{cases}$$

The number of points at which f is continuous, is _____

Question 122 (JAM 2015 • Q40 • ID: JAM_2015_Q40) MSQ +2 M Neg: Nil (0)

Q.10

Let $f:(-1,1) \longrightarrow \mathbb{R}$ be the function defined by

$$f(x) = x^2 e^{1/(1-x^2)}$$

Then

(A) f is decreasing in $(-1,0)$

(B) f is increasing in $(0,1)$

(C) $f(x) = 1$ has two solutions in $(-1,1)$

(D) $f(x) = 1$ has no solutions in $(-1,1)$

Question 123 (JAM 2015 • Q39 • ID: JAM_2015_Q39) MSQ +2 M Neg: Nil (0)

Q.9

Let $f, g : [0,1] \to [0,1]$ be functions. Let $R(f)$ and $R(g)$ be the ranges of f and g, respectively. Which of the following statements is (are) true?

(A) If $f(x) \leq g(x)$ for all $x \in [0,1]$, then $\sup R(f) \leq \inf R(g)$

(B) If $f(x) \leq g(x)$ for some $x \in [0,1]$, then $\inf R(f) \leq \sup R(g)$

(C) If $f(x) \leq g(y)$ for some $x, y \in [0,1]$, then $\inf R(f) \leq \sup R(g)$

(D) If $f(x) \leq g(y)$ for all $x, y \in [0,1]$, then $\sup R(f) \leq \inf R(g)$

Question 124 (JAM 2015 • Q37 • ID: JAM_2015_Q37) MSQ +2 M Neg: Nil (0)

Q.7 Which of the following statements is (are) true on the interval $\left(0, \frac{\pi}{2}\right)$?

(A) $\cos x < \cos(\sin x)$

(B) $\tan x < x$

(C) $\sqrt{1+x} < 1 + \frac{x}{2} - \frac{x^2}{8}$

(D) $\frac{1-x^2}{2} < \ln(2+x)$

Question 125 (JAM 2015 • Q31 • ID: JAM_2015_Q31) MSQ +2 M Neg: Nil (0)

Q.1

Let $f:\mathbb{R}\to\mathbb{R}$ be a function defined by $f(x)=\int_{-5}^{x}(t-1)^3dt$.
In which of the following interval(s), f takes the value 1?

(A) $[-6,\ 0]$ (B) $[-2,\ 4]$ (C) $[2,\ 8]$ (D) $[6,\ 12]$

Question 126 (JAM 2015 • Q28 • ID: JAM_2015_Q28) MCQ +2 M Neg: −0.67

Q.28

For $m,n\in\mathbb{N}$, define $f_{m,n}(x)=\begin{cases}x^m\sin\left(\frac{1}{x^n}\right), & x\neq 0\\ 0 & x=0\end{cases}$
Then at $x=0$, $f_{m,n}$ is

(A) differentiable for each pair m,n with $m>n$

(B) differentiable for each pair m,n with $m<n$

(C) not differentiable for each pair m,n with $m>n$

(D) not differentiable for each pair m,n with $m<n$

Question 127 (JAM 2015 • Q27 • ID: JAM_2015_Q27) MCQ +2 M Neg: −0.67

Q.27

For $n\geq 2$, let $f_n:\mathbb{R}\to\mathbb{R}$ be given by $f_n(x)=x^n\sin x$. Then at $x=0$, f_n has a

(A) local maximum if n is even

(B) local maximum if n is odd

(C) local minimum if n is even

(D) local minimum if n is odd

Question 128 (JAM 2015 • Q21 • ID: JAM_2015_Q21) MCQ +2 M Neg: −0.67

Q.21 Let $f:\mathbb{R}\to\mathbb{R}$ be a strictly increasing continuous function. If $\{a_n\}$ is a sequence in $[0,1]$, then the sequence $\{f(a_n)\}$ is

(A) increasing
(B) bounded
(C) convergent
(D) not necessarily bounded

Question 129 (JAM 2015 • Q17 • ID: JAM_2015_Q17) MCQ +2 M Neg: −0.67

Q.17 The limit

$$\lim_{x\to 0+}\frac{1}{\sin^2 x}\int_{\frac{x}{2}}^{x}\sin^{-1}t\,dt$$

is equal to

(A) 0
(B) $\frac{1}{8}$
(C) $\frac{1}{4}$
(D) $\frac{3}{8}$

Question 130 (JAM 2015 • Q7 • ID: JAM_2015_Q7) MCQ +1 M Neg: −0.33

Q.7 Let $f:\mathbb{R}\to\mathbb{R}$ be a differentiable function with $f(0)=0$. If for all $x\in\mathbb{R}$, $1<f'(x)<2$, then which one of the following statements is true on $(0,\infty)$?

(A) f is unbounded
(B) f is increasing and bounded
(C) f has at least one zero
(D) f is periodic

Question 131 (JAM 2014 • Q33 • ID: JAM_2014_Q33) MCQ +2 M Neg: −0.67

Q.33 Let $f:(0,\infty)\to\mathbb{R}$ be a differentiable function such that $f'(x^2)=1-x^3$ for all $x>0$ and $f(1)=0$. Then $f(4)$ equals

(A) $\frac{-47}{5}$
(B) $\frac{-47}{10}$
(C) $\frac{-16}{5}$
(D) $\frac{-8}{5}$

Question 132 (JAM 2014 • Q26 • ID: JAM_2014_Q26) MCQ +2 M Neg: −0.67

Q.26 Let $f:\mathbb{R}\to\mathbb{R}$ be a function with continuous derivative such that $f(\sqrt{2})=2$ and $f(x)=\lim_{t\to 0}\frac{1}{2t}\int_{x-t}^{x+t} s\,f'(s)\,ds$ for all $x\in\mathbb{R}$. Then $f(3)$ equals

(A) $\sqrt{3}$ (B) $3\sqrt{2}$ (C) $3\sqrt{3}$ (D) 9

Question 133 (JAM 2014 • Q23 • ID: JAM_2014_Q23) MCQ +2 M Neg: −0.67

Q.23 Let $f(x)=x^3+x$ and $g(x)=x^3-x$ for all $x\in\mathbb{R}$. If f^{-1} denotes the inverse function of f, then the derivative of the composite function $g\circ f^{-1}$ at the point 2 is

(A) $\frac{2}{13}$ (B) $\frac{1}{2}$ (C) $\frac{11}{13}$ (D) $\frac{11}{4}$

Question 134 (JAM 2014 • Q22 • ID: JAM_2014_Q22) MCQ +2 M Neg: −0.67

Q.22 Let $S=[0,1]\cup[2,3)$ and let $f:S\to\mathbb{R}$ be defined by $f(x)=\begin{cases}2x & \text{if } x\in[0,1],\\ 8-2x & \text{if } x\in[2,3).\end{cases}$

If $T=\{f(x):x\in S\}$, then the inverse function $f^{-1}:T\to S$

(A) does NOT exist (B) exists and is continuous

(C) exists and is NOT continuous (D) exists and is monotonic

Question 135 (JAM 2014 • Q10 • ID: JAM_2014_Q10) MCQ +1 M Neg: −0.33

Q.10 Let $f:\mathbb{R}\to\mathbb{R}$ be a continuous function satisfying $x+\int_0^x f(t)\,dt=e^x-1$ for all $x\in\mathbb{R}$. Then the set $\{x\in\mathbb{R}:1\le f(x)\le 2\}$ is the interval

(A) $[\log 2,\ \log 3]$ (B) $[2\log 2,\ 3\log 3]$

(C) $[e-1,\ e^2-1]$ (D) $[0,\ e^2]$

Question 136 (JAM 2014 • Q2 • ID: JAM_2014_Q2) MCQ +1 M Neg: −0.33

Q.2 The coefficient of $(x-1)^2$ in the Taylor series expansion of $f(x)=xe^x$ $(x\in\mathbb{R})$ about the point $x=1$ is

(A) $\frac{e}{2}$ (B) $2e$ (C) $\frac{3e}{2}$ (D) $3e$

Question 137 (JAM 2014 • Q1 • ID: JAM_2014_Q1) MCQ +1 M Neg: −0.33

Q.1 Let $f(x)=|x^2-25|$ for all $x\in\mathbb{R}$. The total number of points of $\mathbb{R}$ at which f attains a local extremum (minimum or maximum) is

(A) 1 (B) 2 (C) 3 (D) 4

Question 138 (JAM 2013 • Q5 • ID: JAM_2013_Q5) MCQ +2 M Neg: −0.67

Let the function $f(x)$ be defined by

$$f(x)=\begin{cases}e^x, & x \text{ is rational}\\ e^{1-x}, & x \text{ is irrational}\end{cases}$$

for x in $(0, 1)$. Then

(A) f is continuous at every point in $(0, 1)$

(B) f is discontinuous at every point in $(0, 1)$

(C) f is discontinuous only at one point in $(0, 1)$

(D) f is continuous only at one point in $(0, 1)$

Question 139 (JAM 2013 • Q4 • ID: JAM_2013_Q4) MCQ +2 M Neg: −0.67

Q.4 Let $[x]$ denote the greatest integer function of x. The value of α for which the function

$$f(x)=\begin{cases}\dfrac{\sin[-x^2]}{[-x^2]}, & x\neq 0\\ \alpha, & x=0\end{cases}$$

is continuous at $x=0$ is

(A) 0 (B) $\sin(-1)$ (C) $\sin 1$ (D) 1

Question 140 (JAM 2012 • Q2 • ID: JAM_2012_Q2) MCQ +6 M Neg: −2.0

Q.2 The number of distinct real roots of the equation $x^9 + x^7 + x^5 + x^3 + x + 1 = 0$ is

(A) 1 (B) 3 (C) 5 (D) 9

Question 141 (JAM 2011 • Q2 • ID: JAM_2011_Q2) MCQ +6 M Neg: −2.0

Q.2 The number of real roots of the equation $x^3 + x - 1 = 0$ is

(A) 0 (B) 1 (C) 2 (D) 3

Question 142 (JAM 2010 • Q5 • ID: JAM_2010_Q5) MCQ +6 M Neg: −2.0

Q.5 Let $f : (1,4) \to \mathbf{R}$ be a uniformly continuous function and let $\{a_n\}$ be a Cauchy sequence in $(1, 2)$. Let $x_n = a_n^2 f(a_n^2)$ and $y_n = \frac{1}{1+a_n^2} f(a_n^2)$, for all $n \in \mathbf{N}$. Which of the following statements is true?

(A) Both $\{x_n\}$ and $\{y_n\}$ must be Cauchy sequences in $\mathbf{R}$

(B) $\{x_n\}$ must be a Cauchy sequence in $\mathbf{R}$ but $\{y_n\}$ need not be a Cauchy sequence in $\mathbf{R}$

(C) $\{y_n\}$ must be a Cauchy sequence in $\mathbf{R}$ but $\{x_n\}$ need not be a Cauchy sequence in $\mathbf{R}$

(D) Neither $\{x_n\}$ nor $\{y_n\}$ needs to be a Cauchy sequence in $\mathbf{R}$

Question 143 (JAM 2009 • Q10 • ID: JAM_2009_Q10) MCQ +6 M Neg: −2.0

Q.10 Let a be a non-zero real number. Then $\lim_{x \to a} \frac{1}{x^2 - a^2} \int_a^x \sin(t^2)\,dt$ equals

(A) $\frac{1}{2a}\sin(a^2)$ (B) $\frac{1}{2a}\cos(a^2)$ (C) $-\frac{1}{2a}\sin(a^2)$ (D) $-\frac{1}{2a}\cos(a^2)$

Question 144 (JAM 2009 • Q3 • ID: JAM_2009_Q3) MCQ +6 M Neg: −2.0

Q.3 Which of the following sequences of functions is uniformly convergent on $(0,1)$?

(A) x^n (B) $\frac{n}{nx+1}$ (C) $\frac{x}{nx+1}$ (D) $\frac{1}{nx+1}$

Question 145 (JAM 2008 • Q13 • ID: JAM_2008_Q13) MCQ +6 M Neg: −2.0

Q.13 Let $f:(-1,1)\to\mathbb{R}$ be defined as $f(x)=\frac{x^2}{1-\cos x}$ for $x\neq 0$ and $f(0)=2$. If $f(x)=\sum_{n=0}^{\infty}a_n x^n$ is the Taylor expansion of f for all x in $(-1,1)$, then $\sum_{n=0}^{\infty}a_{2n+1}$ is

(A) 0 (B) 1/2 (C) 1 (D) 2

Question 146 (JAM 2008 • Q8 • ID: JAM_2008_Q8) MCQ +6 M Neg: −2.0

Q.8 Let $f(x)=\int_0^x (x^2-t^2)\,g(t)\,dt$, where g is a real valued continuous function on $\mathbb{R}$. Then $f'(x)$ is equal to

(A) 0 (B) $x^3\,g(x)$ (C) $\int_0^x g(t)\,dt$ (D) $2x\int_0^x g(t)\,dt$

Question 147 (JAM 2008 • Q7 • ID: JAM_2008_Q7) MCQ +6 M Neg: −2.0

Q.7 The set $U=\left\{x\in\mathbb{R}\mid \sin x=\frac{1}{2}\right\}$ is

(A) open (B) closed
(C) both open and closed (D) neither open nor closed

Question 148 (JAM 2007 • Q12 • ID: JAM_2007_Q12) MCQ +6 M Neg: −2.0

12. Let $f:\mathbf{R}\to\mathbf{R}$ be a continuous function. If $\int_0^x f(2t)dt=\frac{x}{\pi}\sin(\pi x)$ for all $x\in\mathbf{R}$, then $f(2)$ is equal to

(A) -1
(B) 0
(C) 1
(D) 2

Question 149 (JAM 2007 • Q8 • ID: JAM_2007_Q8) MCQ +6 M Neg: −2.0

8. For every function $f:[0,1]\to\mathbf{R}$ which is twice differentiable and satisfies $f'(x)\geq 1$ for all $x\in[0,1]$, we must have

(A) $f''(x)\geq 0$ for all $x\in[0,1]$
(B) $f(x)\geq x$ for all $x\in[0,1]$
(C) $f(x_2)-x_2\leq f(x_1)-x_1$ for all $x_1,x_2\in[0,1]$ with $x_2\geq x_1$
(D) $f(x_2)-x_2\geq f(x_1)-x_1$ for all $x_1,x_2\in[0,1]$ with $x_2\geq x_1$

Question 150 (JAM 2007 • Q3 • ID: JAM_2007_Q3) MCQ +6 M Neg: −2.0

3. Which of the following functions is uniformly continuous on the domain as stated?

(A) $f(x)=x^2,\quad x\in\mathbf{R}$
(B) $f(x)=\frac{1}{x},\quad x\in[1,\infty)$
(C) $f(x)=\tan x,\quad x\in(-\pi/2,\pi/2)$
(D) $f(x)=[x],\quad x\in[0,1]$
($[x]$ is the greatest integer less than or equal to x)

Question 151 (JAM 2007 • Q1 • ID: JAM_2007_Q1) MCQ +6 M Neg: −2.0

1. Let $A(t)$ denote the area bounded by the curve $y=e^{-|x|}$, the x-axis and the straight lines $x=-t$ and $x=t$. Then $\lim_{t\to\infty} A(t)$ is equal to

(A) 2
(B) 1
(C) 1/2
(D) 0

Question 152 (JAM 2006 • Q15 • ID: JAM_2006_Q15) MCQ +6 M Neg: −2.0

15. Let $f_n(x)=n\sin^{2n+1}x\cos x$. Then the value of

$$\lim_{n\to\infty}\int_0^{\pi/2} f_n(x)dx - \int_0^{\pi/2}\left(\lim_{n\to\infty} f_n(x)\right)dx$$

is

(A) $1/2$
(B) 0
(C) $-1/2$
(D) $-\infty$

Question 153 (JAM 2006 • Q5 • ID: JAM_2006_Q5) MCQ +6 M Neg: −2.0

5. Let $f(x)=\int_{\sin x}^{\cos x} e^{-t^2}dt$. Then $f'(\pi/4)$ equals

(A) $\sqrt{1/e}$
(B) $-\sqrt{2/e}$
(C) $\sqrt{2/e}$
(D) $-\sqrt{1/e}$

Question 154 (JAM 2006 • Q2 • ID: JAM_2006_Q2) MCQ +6 M Neg: −2.0

2. Let $f(x)=(x-2)^{17}(x+5)^{24}$. Then

(A) f does not have a critical point at 2
(B) f has a minimum at 2
(C) f has a maximum at 2
(D) f has neither a minimum nor a maximum at 2

Question 155 (JAM 2005 • Q15 • ID: JAM_2005_Q15) MCQ +6 M Neg: −2.0

15. Let $a > 1$ and $f, g, h : [-a, a] \longrightarrow \boldsymbol{R}$ be twice differentiable functions such that for some c with $0 < c < 1 < a$,

$f(x) = 0$ only for $x = -a, 0, a$;
$f'(x) = 0 = g(x)$ only $x = -1, 0, 1$;
$g'(x) = 0 = h(x)$ only for $x = -c, c$.

The possible relations between f, g, h are

(A) $f = g'$ and $h = f'$

(B) $f' = g$ and $g' = h$

(C) $f = -g'$ and $h' = g$

(D) $f = -g'$ and $h' = f$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Functions of One Real Variable (2005–2026 Archive) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+2	0.4 to 0.5
2	NAT	+1	−4.0 to −4.0
3	NAT	+1	1.0 to 1.0
4	MSQ	+2	B
5	MSQ	+2	A;C
6	MSQ	+2	A;C;D
7	MCQ	+2	D
8	MCQ	+2	A
9	MCQ	+2	D
10	MCQ	+1	B
11	MCQ	+1	B
12	MCQ	+1	A
13	NAT	+2	9 to 9
14	NAT	+2	1.98 to 2.02
15	NAT	+1	1.98 to 2.02
16	NAT	+1	3.98 to 4.02
17	MSQ	+2	B;C;D
18	MSQ	+2	B;D
19	MCQ	+2	C
20	MCQ	+2	B
21	MCQ	+1	A
22	MCQ	+1	A
23	NAT	+2	1.50 to 1.70
24	NAT	+2	5 to 5
25	NAT	+2	0.24 to 0.26
26	NAT	+1	3 to 3
27	MSQ	+2	B;D
28	MSQ	+2	B;D
29	MCQ	+2	C
30	MCQ	+2	D
31	MCQ	+2	A
32	MCQ	+2	D
33	NAT	+2	0.49 to 0.51
34	NAT	+2	0.74 to 0.76
35	NAT	+1	0.49 to 0.51
36	NAT	+1	0.49 to 0.51
37	MSQ	+2	A, B, D
38	MSQ	+2	A, C, D
53	MCQ	+2	C
54	NAT	+2	0
55	NAT	+2	15
56	NAT	+2	0
57	NAT	+1	2
58	MSQ	+2	B, C, D
59	MSQ	+2	A, C
60	MSQ	+2	A, B
61	MSQ	+2	A, B
62	MCQ	+2	B
63	MCQ	+2	A
64	MCQ	+2	A
65	MCQ	+2	A
66	MCQ	+2	A
67	MCQ	+1	C
68	MCQ	+1	A
69	MCQ	+1	D
70	NAT	+2	0.5 to 0.5
71	NAT	+1	0.08 to 0.09
72	NAT	+1	0 to INF
73	MSQ	+2	A;C;D
74	MSQ	+2	A;C;D
75	MCQ	+2	A
76	MCQ	+2	A
77	MCQ	+1	D
78	MCQ	+1	D
79	NAT	+1	8 to 8
80	NAT	+1	0.25 to 0.25
81	NAT	+1	1 to 1
82	MSQ	+2	B, C
83	MSQ	+2	B, C, D
84	MCQ	+2	A
85	MCQ	+2	D
86	MCQ	+2	C
87	NAT	+2	2.9 to 3.1
88	NAT	+2	0.350 to 0.380
89	NAT	+1	−0.130 to −0.120
105	MCQ	+1	B
106	MCQ	+1	A
107	NAT	+1	0.25 to 0.25
108	NAT	+1	−1 to −1
109	MSQ	+2	A;D
110	MSQ	+2	A;D
111	MCQ	+2	MTA
112	MCQ	+2	A
113	MCQ	+2	C
114	MCQ	+1	B
115	MCQ	+1	A
116	MCQ	+1	A
117	NAT	+2	3
118	NAT	+2	1.41 to 1.42
119	NAT	+2	0.25
120	NAT	+1	3
121	NAT	+1	2
122	MSQ	+2	A;B;C
123	MSQ	+2	B;C;D
124	MSQ	+2	A;D
125	MSQ	+2	A;C;D
126	MCQ	+2	A
127	MCQ	+2	D
128	MCQ	+2	B
129	MCQ	+2	D
130	MCQ	+1	A
131	MCQ	+2	A
132	MCQ	+2	B
133	MCQ	+2	B
134	MCQ	+2	C
135	MCQ	+1	A
136	MCQ	+1	C
137	MCQ	+1	C
138	MCQ	+2	D
139	MCQ	+2	C
140	MCQ	+6	A
141	MCQ	+6	B
142	MCQ	+6	A
143	MCQ	+6	A

Q. No.	Type	Marks	Official Key
39	MCQ	+2	**A**
40	MCQ	+2	**A**
41	MCQ	+2	**A**
42	MCQ	+1	**B**
43	MCQ	+1	**A**
44	NAT	+2	**13.39 to 13.41**
45	NAT	+2	**2.00 to 2.00**
46	NAT	+2	**2.00 to 2.00**
47	NAT	+1	**2.19 to 2.21**
48	MSQ	+2	**A,D**
49	MSQ	+2	**A,B,C**
50	MSQ	+2	**A,B**
51	MCQ	+2	**C**
52	MCQ	+2	**D**

Q. No.	Type	Marks	Official Key
90	MSQ	+2	**B**
91	MCQ	+2	**Marks to All**
92	MCQ	+2	**B**
93	MCQ	+2	**D**
94	MCQ	+2	**A**
95	MCQ	+1	**A**
96	NAT	+2	**0.9 to 1.1**
97	NAT	+2	**0.9 to 1.1**
98	NAT	+1	**54.9 to 55.1**
99	NAT	+1	**1259.9 to 1260.1**
100	MCQ	+2	**C**
101	MCQ	+2	**C**
102	MCQ	+2	**D**
103	MCQ	+1	**D**
104	MCQ	+1	**A**

Q. No.	Type	Marks	Official Key
144	MCQ	+6	**C**
145	MCQ	+6	**A**
146	MCQ	+6	**D**
147	MCQ	+6	**B**
148	MCQ	+6	**A**
149	MCQ	+6	**D**
150	MCQ	+6	**B**
151	MCQ	+6	**A**
152	MCQ	+6	**A**
153	MCQ	+6	**B**
154	MCQ	+6	**D**
155	MCQ	+6	**B**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, −2/3 for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.